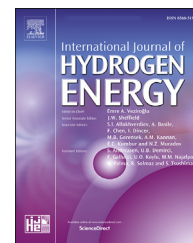


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# Techno-economic analysis of green methanol plant with optimal design of renewable hydrogen production: A case study in China

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## HIGHLIGHTS

- Methanol production from CO<sub>2</sub> and green H<sub>2</sub> based on local weather resources is optimized.
- PV-Wind driven system has the lowest H<sub>2</sub> cost and methanol cost.
- Oxygen selling option and CO<sub>2</sub> capture source obviously affect the methanol cost.
- PV-Wind-H<sub>2</sub>-methanol system can compete with traditional plant considering carbon tax.

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## ABSTRACT

Using renewable energy to convert carbon dioxide (CO<sub>2</sub>) into methanol has gained extensive attention. This study focuses on a green methanol plant which contains both upstream green hydrogen-electricity production system and downstream CO<sub>2</sub>-based methanol synthesis. Three configurations were designed and optimized for the upstream off-grid hydrogen-electricity production system to obtain the minimum hydrogen cost. The optimization was conducted based on hourly meteorological data at Ordos, Inner Mongolia, China, as well as the hydrogen and electricity consumption by the downstream CO<sub>2</sub> hydrogenation unit. On top of that, an integrative techno-economic analysis was conducted on the entire green methanol plant. Furthermore, the effect of key parameters and carbon tax on the economy of such green methanol plant was also explored. The results indicate that (1) for Ordos, the green methanol plant adopting photovoltaic-wind powered hydrogen production system (PV-Wind-H<sub>2</sub>-methanol) with oxygen selling has the lowest leveled cost of methanol, of which the upstream green hydrogen-electricity production system occupies 77%; (2) byproduct oxygen selling and CO<sub>2</sub> source option has a significant effect on the methanol cost; (3) for PV-Wind-H<sub>2</sub>-methanol system, wind turbine cost is the most influential economic factor. To make the methanol cost of the PV-Wind-H<sub>2</sub>-methanol system to be lower than the conventional methanol cost, the cost of wind turbines and PV needs to be reduced by at least 50% at the same time; (4) if

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considering carbon tax, the PV-Wind-H<sub>2</sub>-methanol system will have potential to compete with conventional methanol plant.

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## Introduction

Increasing greenhouse gases in the atmosphere, mostly in the form of CO<sub>2</sub>, are threatening our global climate system. The enormous release of anthropogenic CO<sub>2</sub> is outpacing the nature's carbon cycle causing significant environmental harm such as global warming, ocean acidification, and sea level rise. To mitigate the climate crisis, China, as one of the major emitting countries, has committed to reducing carbon emissions per unit of gross domestic product (GDP) by 60–65% from 2005 levels by 2030 and becoming carbon neutral by 2060.

To reduce the CO<sub>2</sub> emissions, converting CO<sub>2</sub> to value-added products has received significant attention, because CO<sub>2</sub> is a valuable, abundant carbon source that can be converted into fuels and chemicals [1,2]. Methanol (MeOH) is one of the chemicals that can be produced from CO<sub>2</sub> via hydrogenation technology. Methanol is considered as a future energy carrier because it has higher volume-specific energy density compared to other energy carriers such as hydrogen and lithium batteries, and easy to transport. Therefore, the global demand of methanol is expected to rise constantly. These arguments make methanol and its production from CO<sub>2</sub> a hot, intriguing research topic in climate change mitigation.

To obtain greener methanol, the source of hydrogen is important. Although the CO<sub>2</sub> hydrogenation is a CO<sub>2</sub> fixing and utilization process, emissions may occur during the supply of feedstock hydrogen. Therefore, a number of studies focus on the hydrogen production process [3–6]. For instance, Khamhaeng P et al. [5] compared the techno-economic performance of hydrogen production from dehydrogenation and steam reforming of ethanol for CO<sub>2</sub> conversion to methanol. Manhee Byun et al. [4] evaluated the environmental performance of hydrogen production via methanol steam reforming in a membrane reactor and the results show it can lower the CO<sub>2</sub> emission. There are also literatures to obtain methanol with lower carbon emission by improving the hydrogenation process of CO<sub>2</sub> to methanol [7–10]. N. Khunathorncharoenwong et al. [9] proposed a novel alcohol-assisted methanol synthesis process which can increase CO<sub>2</sub> conversion per pass and reduce H<sub>2</sub> consumption. Furthermore, to achieve deeper decarbonization, a considerable amount of research assumes that the hydrogen for CO<sub>2</sub> hydrogenation is produced from renewable energy via water electrolysis [7,11–22]. These studies on green methanol have different concerns, some focusing on process simulation and optimization [7,12,18], and some on techno-economic evaluation [14–16,20,22]. When it comes to economic evaluation, most literatures did their analysis based on a given renewable hydrogen production cost

or renewable electricity price. For instance, Lee et al. [20] performed a techno-economic analysis for renewable methanol synthesis with assumption that the renewable H<sub>2</sub> production cost was 2.3 \$/kgH<sub>2</sub>; Judit Nyári [22] set the renewable electricity price at 0.03 €/kWh; Rivarolo et al. [14] set the price of renewable electricity between 0.03 and 0.05 €/kWh; Cifre et al. [19] assumed the green electricity price as 0.022 €/kWh at peak time and 0.015 €/kWh at off-peak period. In fact, the cost of renewable electricity or green hydrogen varies due to different assumptions, various upstream renewable energy generation configuration, local solar and wind resources and a changing state of technology. Therefore, it's important to consider the effect of these factors on the cost of hydrogen when applying CO<sub>2</sub> hydrogenation in a certain location. In addition, when coupling the renewable energy system with industrial chemical production, there is a contradiction between the continuous stable hydrogen demand of the chemical process system, and the intermittent nature and fluctuation of the renewable energy system. Therefore, it is essential to consider the design and optimization of the upstream renewable power systems for hydrogen production to better match the downstream demand.

There have been several research efforts directed to the upstream optimal design of hydrogen production powered by renewable energy system [21,23–28]. Murat et al. [24] optimized two types of hybrid renewable power generation systems to supply hydrogen for 25 vehicles on a daily basis in Turkey, using Homer software; Abdin et al. [23] considered nine different renewable energy systems to produce hydrogen and simulated the optimum size of system components to identify the cost-effective configurations based on particular locations. Yate et al. [21] reported a Monte Carlo approach to explore a wide range of input assumptions to identify key cost drivers, targets, and localized conditions necessary for competitive stand-alone PV powered hydrogen electrolysis. These studies focus on the sizing optimization of the hybrid energy system components to minimize the cost of hydrogen. Therefore, when it comes to economic evaluation, these studies usually used the hydrogen cost of the hybrid energy system as the final economic index, without considering the downstream hydrogen utilization system. In addition, as an important input condition, the hydrogen load profiles of these studies are mainly estimated based on a hydrogen refueling station or by scaling an annual daily average value considering randomness. Little studies have considered using hydrogen profile from a continuous chemical production process, especially CO<sub>2</sub> hydrogenation to methanol system. Research considering both capacity optimization and economic evaluation for a continuous renewable energy integrated CO<sub>2</sub> hydrogenation to methanol production is still rare in present China.

Therefore, this paper focus on a green CO<sub>2</sub>-based methanol plant which contains upstream green hydrogen-electricity production system and downstream CO<sub>2</sub> hydrogenation. The goal of this study is to explore the optimum capacity configuration of the upstream hybrid energy system to provide green hydrogen and power in areas rich in renewable energy, and evaluate the economic feasibility of the entire green methanol plant, thus providing possible solutions for China to achieve carbon neutrality by 2060. In this study, we proposed a green methanol plant including upstream hybrid energy system to provide stable green hydrogen and electricity, and downstream CO<sub>2</sub> hydrogenation to produce methanol. The capacity configuration of the components in the hybrid energy system was optimized to minimize the upstream system cost, considering hourly weather condition and methanol capacity. With the optimal component size, the economy of the whole system was evaluated by using levelized cost of methanol (LCOM) as the main economic index. Furthermore, sensitivity analysis was conducted to evaluate the effect of key components (e.g. wind turbines, PV arrays, electrolyzers) and carbon tax on the LCOM of the green methanol plant.

## Methodology

### System definition

Fig. 1 presents the green methanol plant we proposed. The green methanol plant was assumed located at Ordos, Inner Mongolia, China (38°30'N, 108°7'E) which is in abundant wind and solar

energy resources. The specific data of renewable energy resources were obtained from the Global Modeling and Assimilation Office (GMAO) [29]. As shown in Fig. 2, the wind speed at Ordos, Inner Mongolia ranges within 3.30–4.86 m/s, at 10 m above the surface of earth, with an average annual wind speed of 4.04 m/s; the solar radiation varies between 2.78 and 6.87 kWh/m<sup>2</sup>/day, with an annual average value of 4.98 kWh/m<sup>2</sup>/day.

In the green methanol plant, the upstream hybrid energy system produces green hydrogen as feedstock to synthesis methanol via CO<sub>2</sub> hydrogenation. The CO<sub>2</sub> is captured from a nearby coal-based methanol plant and delivered based on pipelines. In addition to providing hydrogen, the upstream energy system also provides electricity for the whole green methanol plant. Because chemical production process usually requires a stable feed-in, the hydrogen and electricity load profile in this study is stable, which means the green methanol plant needs to provide the same amount of hydrogen and electricity every hour. The total demand of hydrogen and electricity are dependent on the green methanol plant scale which can be obtained via Aspen plus. The baseline scale of the green methanol plant is assumed 10,000 ton/yr in this study.

### Renewable hydrogen-electricity production systems

#### Description of the green hydrogen-electricity production systems and components

Three types of stand-alone green hydrogen-electricity production systems were considered: Wind-Battery-Hydrogen storage (Wind-H<sub>2</sub>), PV-Battery-Hydrogen storage (PV-H<sub>2</sub>) and Wind-PV-Battery-Hydrogen storage (PV-Wind-H<sub>2</sub>) systems. As

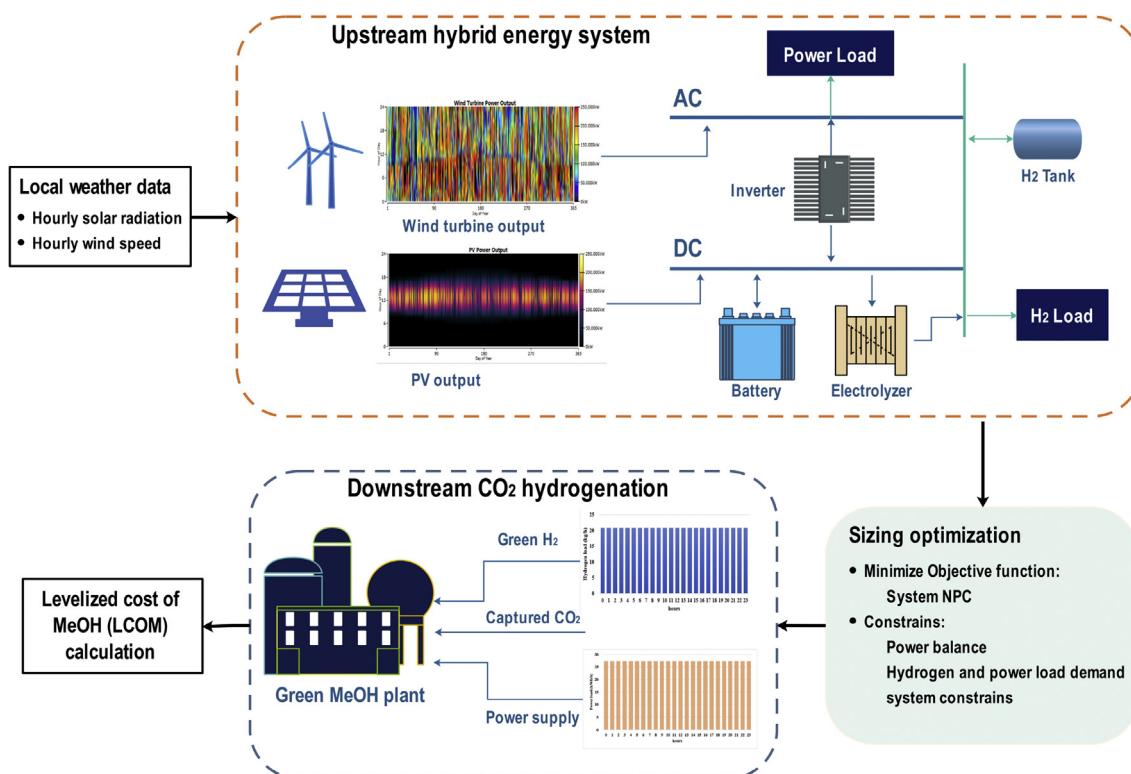


Fig. 1 – Overview of the green Methanol plant via CO<sub>2</sub> hydrogenation. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

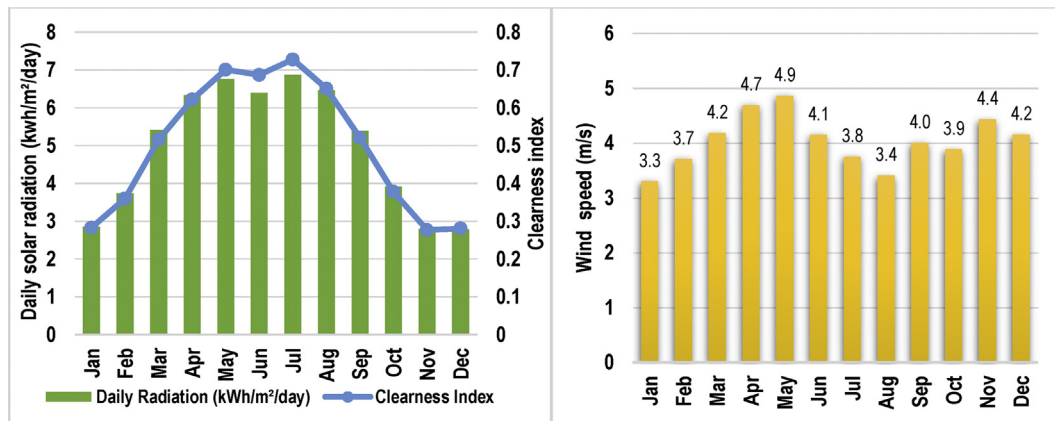


Fig. 2 – Monthly average daily solar radiation and wind speed at ordos.

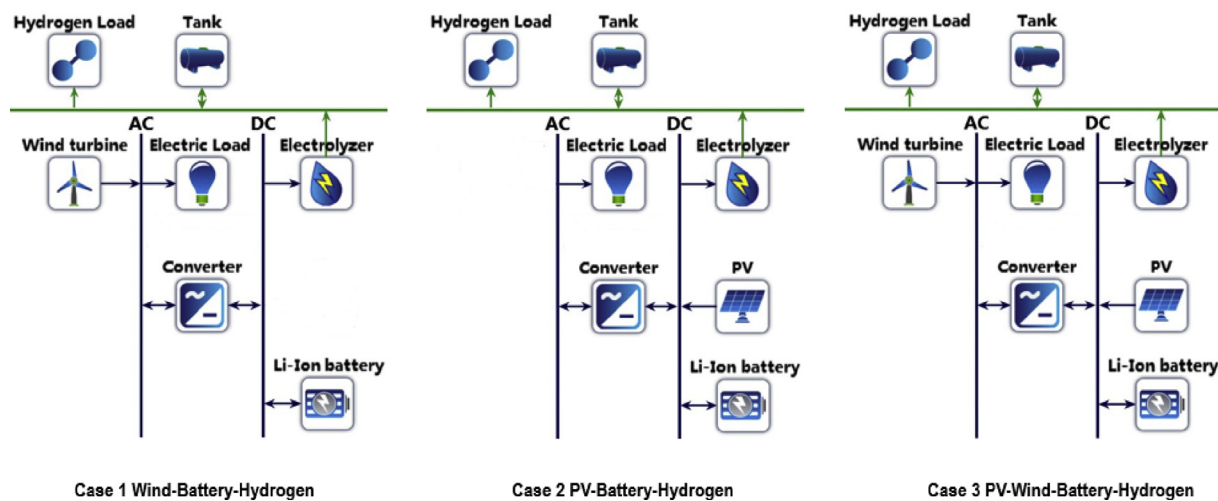


Fig. 3 – Hybrid energy systems for green hydrogen production.

shown in Fig. 3. Wind-H<sub>2</sub> and PV-H<sub>2</sub> systems are solely wind or solar energy driven power systems. To analyze the feasibility of wind-solar complementary generation system, PV-Wind-H<sub>2</sub> system combines wind and solar energy for power generation. These systems deal with the combination of PV array, wind turbine, battery, converter, water electrolysis unit, and hydrogen storage tank. The detail techno-economic parameters used for the green hydrogen-electricity production system is presented in Table 1. Detailed description of PV array, wind turbine, battery, converter, water electrolysis unit, and hydrogen storage tank are available in the supplement material.

#### Optimal sizing of the green hydrogen-electricity production systems

The optimal sizing of each system configuration was determined with aid of the Homer software which is a commercial tool developed by U.S. National Renewable Energy Laboratory (NREL). Fig. 4 illustrates the steps of the optimization process.

The input data for simulation and optimization include meteorological data, load profile, technical and economic data of equipment, and search space. After the input data are fed into Homer, each plan from the search space is simulated. The objective function of each plan is the net present cost (NPC) which is the present value of the sum of costs minus the sum of revenues. The cost covers the initial capital investment, replacement cost, and operating & maintenance cost of the equipment. The NPC of feasible plans is calculated and the operating results of the designed system in each time step (8760 h) was output. Feasibility means the hydrogen and power demand of each time is supplied by the system and the power balance constraint is satisfied in each time step. In our study, a 10% operating reserve load in current time step was set, which means the system must keep enough spare capacity operating to serve a sudden 10% increase in the primary load. At the end, the feasible plans are ranked based on the minimum NPC so that the plan with minimum NPC is



**Table 1 – Techno-economic data for the upstream hybrid energy system.**

| System component                                     |       |           |         |
|------------------------------------------------------|-------|-----------|---------|
| <b>PV unit</b>                                       |       |           |         |
| Lifetime                                             | 25    | Year      | [30]    |
| Derating factor                                      | 90    | %         |         |
| Ground reflection                                    | 20    | %         |         |
| Capital cost                                         | 3500  | RMB/kW    | [31–33] |
| Operating and maintenance cost                       | 40    | RMB/kW/yr | [31–33] |
| <b>Wind turbine system</b>                           |       |           |         |
| Rated power                                          | 4200  | kW        |         |
| Lifetime                                             | 20    | Year      | [24]    |
| Hub height                                           | 122   | m         |         |
| Rotor diameter                                       | 115.7 | m         |         |
| Cut-out wind speed                                   | 28–34 | m/s       |         |
| Capital cost                                         | 7000  | RMB/kW    | [34–36] |
| Operating and maintenance cost                       | 32    | RMB/kW/yr | [34–36] |
| <b>Batteries (Generic 1 kWh lithium ion battery)</b> |       |           |         |
| Nominal voltage                                      | 6     | V         |         |
| Nominal capacity                                     | 1     | kWh       |         |
| Maximum capacity                                     | 167   | Ah        |         |
| Roundtrip efficiency                                 | 90    | %         |         |
| Maximum charge current                               | 167   | A         |         |
| Minimum discharge current                            | 500   | %         |         |
| Lifetime                                             | 10    | Year      | [37,38] |
| Capital cost                                         | 1200  | RMB/kW    | [37,38] |
| Operating and maintenance cost                       | 12    | RMB/kW/yr | [37,38] |
| <b>Converter</b>                                     |       |           |         |
| Lifetime                                             | 15    | Year      | [24]    |
| Efficiency                                           | 95    | %         |         |
| Capital cost                                         | 800   | RMB/kW    | [39]    |
| Operating and maintenance cost                       | 8     | RMB/kW/yr | [39]    |
| <b>PEM electrolyzer</b>                              |       |           |         |
| Lifetime                                             | 7     | Year      | [40]    |
| Efficiency                                           | 80    | %         |         |
| Capital cost                                         | 4600  | RMB/kW    | [40]    |
| Operating and maintenance cost                       | 46    | RMB/kW/yr | [40]    |
| <b>Storage tank</b>                                  |       |           |         |
| Lifetime                                             | 20    | Year      | [30]    |
| Capital cost                                         | 6807  | RMB/kW    | [41]    |
| Operating and maintenance cost                       | 6.8   | RMB/kW/yr | [41]    |

considered as the best plan. Detailed input data and optimization constrains we used are available in the supplement material.

### Methanol synthesis via CO<sub>2</sub> hydrogenation

Methanol is produced through the catalytic hydrogenation of CO<sub>2</sub>. In the presence of Cu/ZnO/Al<sub>2</sub>O<sub>3</sub> catalyst, CO<sub>2</sub> reacts with H<sub>2</sub> to form methanol at 50–100 bar, 210–270 °C, and a H<sub>2</sub>:CO<sub>2</sub> molar ratio of 3:1 [11,12,17,42]. The CO<sub>2</sub> was assumed captured from the gas purification unit in a coal-based methanol plant [43]. The purify of the CO<sub>2</sub> flow out of the gas cleaning unit (Recticol) from the conventional methanol plant can reach 99.5% [44–46], which is enough for industrial application. We choose pipelines to transport CO<sub>2</sub>, which are widely used and economically preferable [47–49]. CO<sub>2</sub> is transported in supercritical phase at pressure 110 bar [47]. The transportation distance for CO<sub>2</sub> is assumed to be 100 km and the pressure drop from transportation of CO<sub>2</sub> is 0.1–0.2 bar/km [47,50]. Therefore, we assume that there is no recompression CO<sub>2</sub> as the pressure after 100 km of transportation is about 90 bar,

which is higher than 50 bar required for hydrogenation of CO<sub>2</sub> to produce methanol. As the hydrogen is immediately used for methanol synthesis when it is produced, the pressure of feed stream H<sub>2</sub> for the downstream process is assumed 30 bar based on the output pressure of a PEM electrolyzer [51]. Detailed mass and energy balance of the methanol production were obtained by Aspen simulation.

Fig. 5 shows the process flowsheet of the CO<sub>2</sub>-based methanol synthesis process modeled by Aspen Plus. The whole process was simulated using Redlich–Kwong–Soave equation of state (RK-SOAVE) and non-random two-liquid model (NRTL) property method, based on literature recommendations for such systems [18,22]. The synthetic raw material gas CO<sub>2</sub> (14 °C, 90 bar) and H<sub>2</sub> (50 °C, 30 bar), mixed with circulating gas, is compressed to about 50 bar by circulating compressor (REPUMP), and heated to about 250 °C by heat exchanger (HX1) before entering the reactor (REACTOR). The reaction temperature of CO<sub>2</sub> hydrogenation is 250 °C, and the reaction pressure is 50 bar. The reaction product is cooled to 30 °C by circulating water before entering gas-liquid separation tower (SEP1). Most of the obtained gas is recycled back to the reactor, and the rest is used as gallop. Refined methanol is obtained from methanol-rich solution after two-column distillation system (DIST1+DIST2). Steam is generated from the combustion with air of the purge (PURGE) and the gaseous streams S7 and VAPOR. Detailed simulation information is provided in the supplement material.

### Integrative economic analysis

#### Levelized cost of methanol (LCOM)

The levelized cost of methanol (LCOM, in RMB/ton) of the target system is selected as the main indicator to evaluate the economy of each case. It was calculated by dividing the total annualized cost of the system ( $C_{ann,tot}$ , RMB/yr) by the annual methanol output ( $M_{MeOH}$ , ton/yr) (Eq. (1)).

$$LCOM = \frac{C_{ann,tot}}{M_{MeOH}} \quad (1)$$

The total annualized cost of the system ( $C_{ann,tot}$ ) was calculated by multiplying the total net present cost ( $C_{NPC,tot}$ , RMB) and capital recovery factor (CRF) (Eq. (2)) [23,24].

$$C_{ann,tot} = C_{NPC,tot} * CRF \quad (2)$$

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (3)$$

$$i = \frac{i' - f}{1 + f} \quad (4)$$

where  $C_{NPC,tot}$  the present value of all the costs of installing and operating the components over the project period, minus the present value of all the revenues of a system. The capital recovery factor (CRF) is a ratio used to calculate the present value of an annuity (a series of equal annual cash flows);  $n$  is the project lifetime;  $i$  denotes real discount rate;  $i'$  is the nominal discount rate;  $f$  is the expected inflation rate.

The total NPC cost ( $C_{NPC,tot}$ ) over the project period is calculated by Eq. (5) [52].

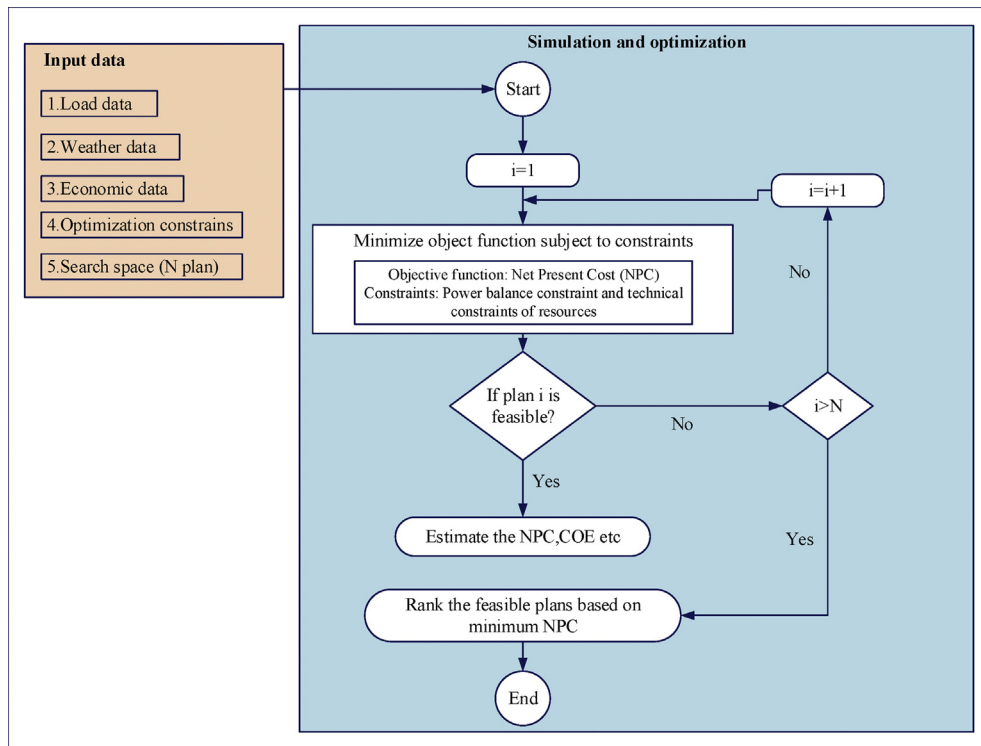


Fig. 4 – Framework of optimization procedure in HOMER.

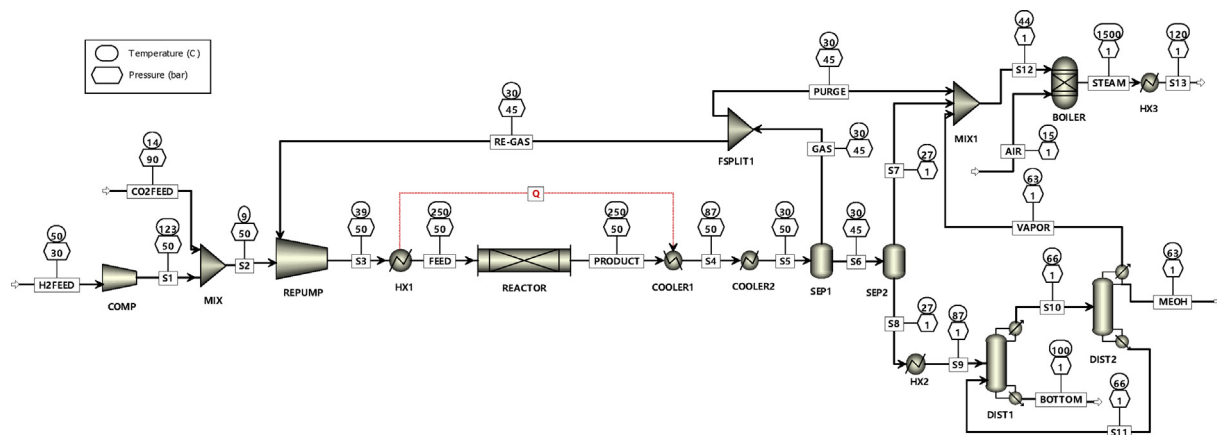


Fig. 5 – Aspen flowsheet of CO<sub>2</sub> hydrogenation.

$$C_{NPC,tot} = TCC + \sum_{j=1}^n C_{NPC,j} \quad (5)$$

where  $TCC$  is the total initial capital cost of the project at year zero, RMB;  $n$  is the project life time;  $C_{NPC,j}$  is the net present cost of the project at year  $j$ , RMB/yr.  $TCC$  is estimated using Eq. (6)

$$TCC = \left(1 + \sum_{i=1}^n RF_i\right) \times TPU \quad (6)$$

where  $TPU$  is the total capital investment of process units;  $RF_i$

is the ratio factors for indirect costs including general facilities, engineering permitting and startup, contingencies and working capital, land and misc (see Table 2).  $TPU$  comes from the upstream hybrid energy system and the downstream CO<sub>2</sub> hydrogenation. The capital investment of units in the upstream hybrid energy system was calculated based on the capacity of each component and its unit capital cost. The capacity of each component was obtained via Homer optimization and the unit capital cost of PV arrays, wind turbines batteries, converter etc. which were taken from current

**Table 2 – Key economic assumptions used in this study.**

| Item                                                                  | Value                                                                                                         | Unit                   |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------|
| <b>1) General economic assumption [51,59]</b>                         |                                                                                                               |                        |
| Baseline green methanol plant capacity                                | 10,000                                                                                                        | ton/yr                 |
| Operating hours                                                       | 8400                                                                                                          | hours                  |
| Nominal discounted rate [24,44]                                       | 8%                                                                                                            | —                      |
| Inflation rate [60]                                                   | 2%                                                                                                            | —                      |
| Project period, n                                                     | 20                                                                                                            | years                  |
| Exchange rate of RMB                                                  | 6.8                                                                                                           | RMB ¥ : US\$           |
| Exchange rate of RMB                                                  | 7.73                                                                                                          | RMB ¥ : Euro€          |
| Life time for methanol production                                     | 20                                                                                                            | years                  |
| General facilities                                                    | 10%                                                                                                           | of TPU                 |
| Engineering permitting and startup                                    | 10%                                                                                                           | of TPU                 |
| Contingencies                                                         | 5%                                                                                                            | of TPU                 |
| Working capital, land and misc                                        | 3%                                                                                                            | of TPU                 |
| Non-fuel operating cost                                               | 0.5%                                                                                                          | of TCC                 |
| <b>2) Cost estimation for methanol reactor, compressor and boiler</b> |                                                                                                               |                        |
| Methanol reactor [61]                                                 | $C_{MeOH} = 10.9 \cdot 10^7 \cdot \left( \frac{M_{gas \text{ entering reactor, kg/h}}}{54000} \right)^{0.65}$ | RMB                    |
| Compressor [59]                                                       | $C_{compressor} = 2.72 \cdot 10^5 \cdot P(kW)^{0.6038}$                                                       | RMB                    |
| Boiler [22]                                                           | $C_{boiler} = 3.155 \cdot 10^7 \cdot \left( \frac{Heat \text{ duty}(MW_{th})}{55.6} \right)^{0.6}$            | RMB                    |
| <b>3) Price of raw material, utilities and by-products [47,55,59]</b> |                                                                                                               |                        |
| CO <sub>2</sub> from coal-based methanol plant                        | 50                                                                                                            | RMB/tonCO <sub>2</sub> |
| CO <sub>2</sub> from PC super-critical                                | 408                                                                                                           | RMB/tonCO <sub>2</sub> |
| CO <sub>2</sub> from IGCC                                             | 550                                                                                                           | RMB/tonCO <sub>2</sub> |
| CO <sub>2</sub> from NGCC                                             | 673                                                                                                           | RMB/tonCO <sub>2</sub> |
| CO <sub>2</sub> from iron & steel industry                            | 503                                                                                                           | RMB/tonCO <sub>2</sub> |
| CO <sub>2</sub> from fertilizer industry                              | 189                                                                                                           | RMB/tonCO <sub>2</sub> |
| Fresh water                                                           | 1.50                                                                                                          | RMB/ton                |
| Cooling water                                                         | 0.30                                                                                                          | RMB/ton                |
| Steam                                                                 | 100                                                                                                           | RMB/ton                |
| Catalyst                                                              | 70                                                                                                            | RMB/kg                 |
| Oxygen                                                                | 1000                                                                                                          | RMB/ton                |

market bidding price or literature, listed in Table 1. The capital investments for equipment in the downstream CO<sub>2</sub> hydrogenation system excluding the methanol reactor, compressors and boiler, were estimated using Aspen Plus Economic Analyzer. The capital investment related to the methanol synthesis reactor, compressors and boiler were estimated with reference to literatures, listed in Table 2.

$C_{NPC,j}$  is the net present cost of the project at year  $j$ , determined using Eq. (7).

$$C_{NPC,j} = (C_{rep} + C_{O\&M} + C_{sav}) / (1 + i)^j \quad (7)$$

where  $C_{rep}$ ,  $C_{O\&M}$ , and  $C_{sav}$  represent the replacement cost, salvage, operating and maintaining (O&M) cost in RMB at the year  $j$ ; the replacement cost ( $C_{rep}$ ) is the cost of replacing a component at the end of its lifetime, which is set at 80% of the its initial capital cost; the salvage value ( $C_{sav}$ ) is the value remaining in a component of the power system at the end of the project lifetime, which is assumed directly proportional to its remaining life; the O&M cost ( $C_{O\&M}$ ) covers the cost of feedstock for production, utilities, by-products which are negative sold out at the market price and other non-fuel O&M cost which is assumed 0.5% of the TCC.

$$C_{O\&M} = C_{feed} + C_{utilities} + C_{byproduct} + C_{non-fuel \text{ O\&M}} \quad (8)$$

CO<sub>2</sub> capture cost includes capture, compression and transportation cost. In this study, the CO<sub>2</sub> is assumed from a

conventional coal-based methanol plant and the CO<sub>2</sub> capture cost is assumed to be 0 RMB/tonCO<sub>2</sub> as this is essentially a waste product from syngas cleaning unit and therefore, the cost for separating CO<sub>2</sub> is assigned to coal-based methanol cost [51].

The captured CO<sub>2</sub> (3 bar, 26 °C) must be compressed to meet the requirements of transportation. The pipeline transportation used in this study usually adopts the supercritical state (pressure>73.8 bar, temperature>31.1 °C), because the density of CO<sub>2</sub> in the supercritical state is close to liquid and close to viscous gas. Therefore, the compressed CO<sub>2</sub> gas is assumed to be 150 bar, 30–40 °C. The investment cost of the compressor station can be estimated using the investment estimation formula of the compressor station used by the European Research Center of the International Energy Agency (IEA) for pipeline transportation of CO<sub>2</sub> [53,54].

$$I_{booster} = [(2.3W + 0.15) \times 10^6] \times r \quad (9)$$

where  $I_{booster}$  is the investment cost of compressor station;  $W$  is the net power of compressor, MW;  $r$  is the exchange rate of RMB against USD.

The total compression cost was estimated as 40 RMB/ton under the baseline case, including equipment depreciation and operating expenses [53,54]. The pipeline delivery cost of CO<sub>2</sub> was estimated as 10 RMB/ton based on industrial transportation scale and 100 km using Rubin's model and Gao's model [14,47]. Accordingly, the total CO<sub>2</sub> cost from coal-based

**Table 3 – Key performance indicators and consumption for the proposed methanol synthesis.**

| Item                                          | Value    | Unit         |
|-----------------------------------------------|----------|--------------|
| Methanol output                               | 10003.97 | ton/year     |
| Methanol purity                               | 99.92    | wt. %        |
| CO <sub>2</sub> usage per ton methanol        | 1.59     | ton/ton-MeOH |
| H <sub>2</sub> usage per ton methanol         | 0.22     | ton/ton-MeOH |
| Hourly H <sub>2</sub> load of upstream system | 259      | kg/hour      |
| Recycle ratio                                 | 5.5      | mol/mol      |
| CO <sub>2</sub> conversion per pass           | 25%      | mol %        |
| Overall CO <sub>2</sub> conversion            | 96.37    | mol %        |
| Power of H <sub>2</sub> feed (COMP)           | 77.22    | kW           |
| Power of recycle compressor (REPUMP)          | 115.17   | kW           |
| Heat generated in the reactor                 | −688.13  | kW           |
| Heat duty of HX2                              | 1020.32  | kW           |
| Heat duty of reboiler (DIST1)                 | 281.22   | kW           |
| Heat duty of reboiler (DIST2)                 | 204.30   | kW           |
| Heat generated of the boiler                  | −505.21  | kW           |
| Electricity usage                             | 161.75   | kWh/ton-MeOH |
| Hourly electricity load of upstream system    | 192      | kWh/hour     |
| Steam usage                                   | 0.43     | ton/ton-MeOH |
| Cooling water usage                           | 618.77   | ton/ton-MeOH |

methanol plant is 50 RMB/ton by adding up capture, compression and transportation cost. Detailed estimation of the CO<sub>2</sub> cost can be found in the [supplement material](#).

Captured CO<sub>2</sub> cost differs for different capture sources. For comparison, CO<sub>2</sub> captured from pulverized-coal (PC) supercritical power plant, integrated gasification combined cycle (IGCC) plant, natural gas combined cycle (NGCC) power plant, fertilizer and iron-making industry was also considered. The CO<sub>2</sub> cost from these sources refers to a CCS study report published by Global CCS Institute [55]. The cost estimates presented in this report include CO<sub>2</sub> capture and compression, pipeline transportation, and storage. Note that since CO<sub>2</sub> from different sources is transported by pipeline after purification, that is, the CO<sub>2</sub> entering the transport stage is product gas with same qualified purity, so the flowrate of CO<sub>2</sub> from different sources is considered to be consistent for a fixed-scale methanol plant.

**Selling price of byproduct oxygen** is highly dependent on its penetration and application in the local market. In this case, the price of oxygen is assumed 1000 RMB/ton which is based on the reported price of industrial oxygen [56–58]. The purity of oxygen out of electrolyzer is more than 99.9%, which is sufficient for industrial applications, therefore, no further purification is needed [15]. [Table 2](#) presents the major economic parameters used in our analyzed systems.

#### Levelized cost of hydrogen (LCOH) and electricity (LCOE)

In addition to the LCOM, the levelized cost of energy (LCOE) and hydrogen (LCOH) of the upstream hybrid energy systems were also calculated, function as assessment indicators for different configurations of the hybrid energy system. The LCOE is defined as the average cost per kWh of useful electrical energy produced by the hybrid energy system, calculated as Eq. (10).

**Table 4 – Optimal sizing of each hybrid energy system.**

|                           | Wind-H <sub>2</sub> | PV-H <sub>2</sub> | PV-Wind-H <sub>2</sub> |
|---------------------------|---------------------|-------------------|------------------------|
| PV (kW)                   | —                   | 96,518            | 16,096                 |
| Wind turbine (kW)         | 33,600              | —                 | 33,600                 |
| Batteries                 | 2769                | 4772              | 814                    |
| Electrolyzer (kW)         | 24,000              | 45,000            | 23,000                 |
| Hydrogen Tank             | 17,000              | 12,000            | 15,000                 |
| Converter (kW)            | 25,521              | 3329              | 24,305                 |
| PV electricity (MWh/yr)   | —                   | 195,059           | 32,529                 |
| Wind electricity (MWh/yr) | 155,978             | —                 | 116,984                |
| Excess electricity (%)    | 23                  | 42                | 21                     |
| Total NPC (million RMB)   | 614                 | 869               | 585                    |

$$LCOE = (C_{ann,upstream} - C_{boiler}H_{served}) / E_{el} \quad (10)$$

where  $C_{ann,upstream}$  is the total annualized cost of the upstream hydrogen-electricity production system, RMB/yr;  $C_{boiler}H_{served}$  means the cost of serving the thermal load which is not included in our study because in the proposed upstream hydrogen-electricity production system, a low-temperature PEM electrolysis system is used, so there is no thermal load from boiler, namely, the annual thermal load ( $H_{served}$ ) is not considered;  $E_{el}$  is the sum of the AC electricity served for the whole green methanol plant and the DC electricity consumed by water electrolysis unit, kWh/yr.

LCOH, also called the hydrogen supply cost, is defined as the annualized cost of the upstream hybrid energy system dividing the total hydrogen production, RMB/kg.

$$LCOH = C_{ann,upstream} / M_{hydrogen} \quad (11)$$

where  $M_{hydrogen}$  denotes the whole hydrogen supply per year, kg/yr.

#### Carbon emission

The total emissions of CO<sub>2</sub> of the methanol plant are calculated based on the direct and indirect emissions as shown in Eq. (12) [45].

$$E_{CO_2,total} = E_{CO_2,direct} + E_{CO_2,indirect} = E_{CO_2,direct} + \sum_{j=1}^n m_j \times EF_{j,CO_2} \quad (12)$$

where  $E_{i,total}$  represent the total amount of CO<sub>2</sub> emission at the methanol production stage,  $E_{i,direct}$  and  $E_{i,indirect}$  represents the direct and indirect emission of CO<sub>2</sub>. Direct carbon emission refers to carbon emitted directly during the production process, contained in the exhaust gas; indirect carbon emission refers to the hidden carbon emissions embodied in intermediates during production, specifically outsourcing utilities (electricity, fresh water, oxygen and compressed air etc).  $m_j$  is the consumption of  $j$ , including energy (e.g. diesel, gasoline) and utilities (e.g. electricity, steam).  $EF_{j,CO_2}$  is the CO<sub>2</sub> emission factor for  $j$ .  $\sum_{s=1}^n m_{s,j} \times EF_{j,CO_2}$  is the aggregate indirect amount of CO<sub>2</sub> emission involved in this stage. The carbon emission factors (see supplement material) of main materials consumed in this



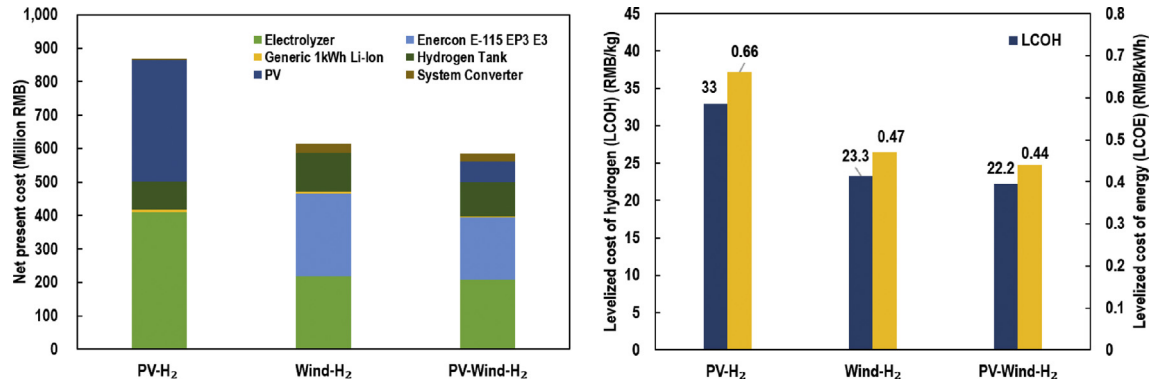


Fig. 6 – Cost summary of upstream hybrid energy system for different configurations.

analysis refer to 2006 IPCC Guidelines for national greenhouse gas inventories and China energy statistics yearbook 2019 released by National bureau of statistics [62,63].

## Results and discussion

### Hydrogen and electricity consumption of the methanol production unit

The simulation of downstream methanol synthesis provides specific hydrogen and electricity load for the upstream green hydrogen-electricity production system. Table 3 presents the key technical indicators for the process of methanol synthesis via

CO<sub>2</sub> hydrogenation (Detailed stream and energy balance data are available in the supplement material). The desired purity of methanol (>99.85%, Grade AA methanol) was achieved after distillation. The CO<sub>2</sub> conversion per pass in this study is 25%, consistent with experimental data from An et al. [14] which developed a highly active Cu/Zn/Al/Zr fibrous catalyst for methanol synthesis via CO<sub>2</sub> hydrogenation. The overall CO<sub>2</sub> conversion rate of the plant is about 96%, which is acceptable compared to other similar simulations ranging in 92%–99% [12,17,64]. The average unit consumption of CO<sub>2</sub> and H<sub>2</sub> is 1.59 and 0.22 ton/ton-MeOH, respectively. To produce 10,000 tons methanol annually, the upstream hybrid energy system needs to provide at least 259 kg hydrogen and 192 kWh electricity hourly. In terms of thermal balance, in the methanol plant, the heat of CO<sub>2</sub>

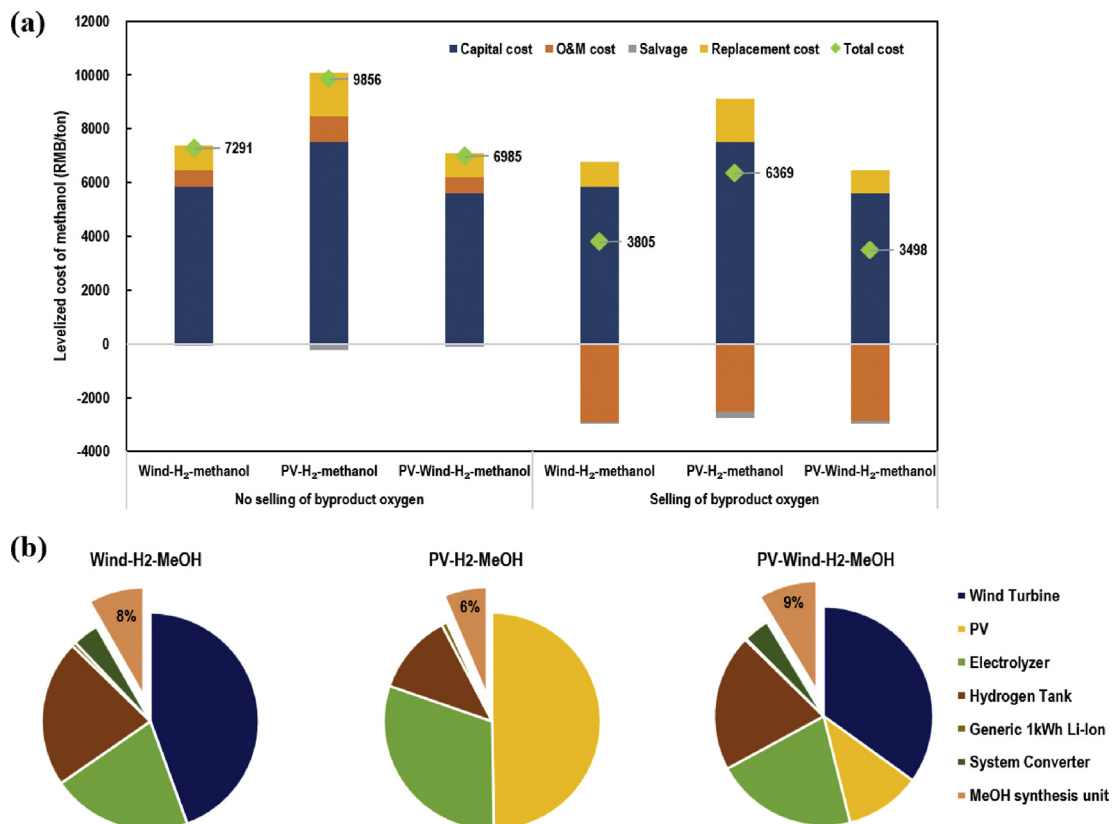


Fig. 7 – (a) Levelized cost of methanol and the cost structure; (b) Initial capital cost categorized by components for all cases.

**Table 5 – Estimation of total capital cost for 10 kt/yr green methanol plant (Million RMB).**

|                                        | Wind-H <sub>2</sub> -methanol | PV-H <sub>2</sub> -methanol | PV-Wind-H <sub>2</sub> -methanol |
|----------------------------------------|-------------------------------|-----------------------------|----------------------------------|
| 1. Direct capital cost                 |                               |                             |                                  |
| 1.1 Hybrid energy                      |                               |                             |                                  |
| Wind turbine                           | 235.20                        | —                           | 176.40                           |
| PV array                               | —                             | 337.81                      | 56.34                            |
| Electrolyzer                           | 110.40                        | 207.00                      | 105.80                           |
| Hydrogen tank                          | 115.72                        | 81.68                       | 102.11                           |
| Battery                                | 3.32                          | 5.73                        | 0.98                             |
| System converter                       | 20.42                         | 2.66                        | 19.44                            |
| 1.2 Methanol synthesis unit            |                               |                             |                                  |
| COOLER1                                | 0.09                          | 0.09                        | 0.09                             |
| COOLER2                                | 0.10                          | 0.10                        | 0.10                             |
| HX1                                    | 0.17                          | 0.17                        | 0.17                             |
| HX2                                    | 0.09                          | 0.09                        | 0.09                             |
| REPUMP                                 | 4.78                          | 4.78                        | 4.78                             |
| COMP                                   | 3.76                          | 3.76                        | 3.76                             |
| REACTOR                                | 29.82                         | 29.82                       | 29.82                            |
| DIST1                                  | 1.36                          | 1.36                        | 1.36                             |
| DIST2                                  | 1.31                          | 1.31                        | 1.31                             |
| SEP1                                   | 0.15                          | 0.15                        | 0.15                             |
| SEP2                                   | 0.16                          | 0.16                        | 0.16                             |
| BOILER                                 | 1.88                          | 1.88                        | 1.88                             |
| Total direct cost                      | 528.73                        | 678.55                      | 504.74                           |
| 2. Indirect capital cost               |                               |                             |                                  |
| 2.1 General facilities                 | 52.87                         | 67.85                       | 50.47                            |
| 2.2 Engineering permitting and startup | 52.87                         | 67.85                       | 50.47                            |
| 2.3 Contingencies                      | 26.44                         | 33.92                       | 25.24                            |
| 2.4 Working capital, land and misc     | 15.86                         | 20.36                       | 15.14                            |
| Total indirect capital cost            | 148.04                        | 189.99                      | 141.32                           |
| 3. Total capital cost (TCC)            | 676.77                        | 868.54                      | 646.07                           |

hydrogenation generated in the reactor can be used to generate high pressure steam. In addition, the purge gases were mixed together to generate steam to provide heat for distillation reboilers and heaters. After heat generation, the plant also needs 0.43 ton/ton-MeOH outsourcing steam to meet the heat demand.

#### Optimal sizing for hybrid energy systems

The optimum results of the upstream hybrid energy systems for methanol plant with different configurations are presented in Table 4 and Fig. 6. The PV-Wind-H<sub>2</sub> system has the lowest system NPC (585 million RMB), LCOH (22.2 RMB/kg) and LCOE (0.44 RMB/kWh), followed by Wind-H<sub>2</sub> system. This is because by combination of PV-Wind, the stability of upstream electricity is increased, leading to a reduction in the required capacity of other components such as electrolyzer and storage tank. The economic results show the system cost of PV-Wind-H<sub>2</sub> system is lower than that of PV-H<sub>2</sub> and Wind-H<sub>2</sub> system, which indicates that the increased cost of introducing photovoltaics is offset by the decreased cost of other components. In contrast, the PV-H<sub>2</sub> system has the highest NPC (869 million RMB), LCOH (33 RMB/kg) and LCOE (0.66 RMB/kWh) because PV-H<sub>2</sub> system utilizes only solar energy which only generates electricity during the day, leading to increasing system capacity and capital investment. Note that the NPC here is the net present cost of the upstream hybrid energy system without considering the downstream methanol synthesis system.

In terms of electricity production, Wind-H<sub>2</sub> system produces 155,978 MWh/yr with 23% surplus electricity while PV-H<sub>2</sub>

system produces 195,059 MWh/yr with 42% excess electricity, due to its much higher system capacity to meet 24-h hydrogen and power demand. PV-Wind-H<sub>2</sub> system has the least excess electricity ratio (21%) with an annual output of 149,513 MWh/yr. 78% of its total electricity is generated by the wind turbines whereas the remaining 22% is generated by the PV arrays. Each case produces more electricity than required. In general, the excess electricity is inevitable because the methanol plant requires stable hydrogen and electricity profile load. It is difficult to reduce the excess electricity by decreasing the size of the power generation system because this will reduce the reliability of the hydrogen and electricity supply to the methanol plant.

#### Economic results of the whole green methanol plant

The levelized cost of methanol (LCOM, RMB/ton) of the target system is selected as the economic indicator to compare different systems. Based on the material balance of the whole system and the optimization results in section [Optimal sizing for hybrid energy systems](#), the LCOM of the whole green methanol plant can be obtained. Fig. 7(a) presents the LCOM for green methanol plant with different upstream hydrogen production systems. In all cases the estimated LCOM is above the conventional coal-based methanol cost, estimated as 1876 RMB/ton-MeOH based on 500 RMB/ton coal price [1,45]. The LCOM as expected decreases significantly with the decreasing capital investment of green hydrogen production system. PV-Wind-H<sub>2</sub> system has the lowest initial investment and correspondingly the system composed of it (PV-Wind-H<sub>2</sub>-methanol) has the

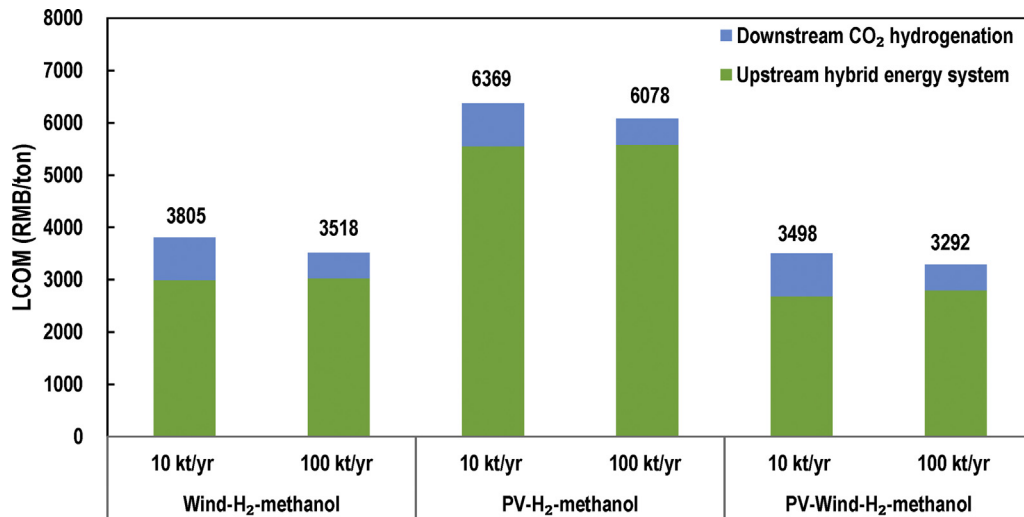


Fig. 8 – Effect of methanol plant scale on the LCOM.

lowest methanol cost. In addition, Fig. 7(a) also shows how LCOM changes when oxygen, the by-product from H<sub>2</sub> production is sold. The green methanol plant adopting PV-Wind-H<sub>2</sub> system with oxygen selling has the lowest LCOM (3498 RMB/ton-MeOH). In all cases significant reduction, around 3487 RMB/ton in LCOM, can be noticed when selling oxygen at 1000 RMB/ton, therefore, considering selling by-product oxygen is a viable option to achieve a better business case.

In terms of the cost structure, the LCOM can be divided into annualized capital cost, operating & management (O&M) cost, replacement cost and salvage. As shown in Fig. 7(a), the capital cost is the major economic contributor for all three cases, followed by replacement cost. 6–9% of the initial capital cost comes from the downstream CO<sub>2</sub> hydrogenation unit and the remaining 91–94% from the upstream green hydrogen-electricity production system (see Table 5 and Fig. 7(b)). The capital cost of the hybrid energy system is most affected by power generation components (wind turbine or PV arrays accounting for more than 45% of the total capital cost), followed by electrolyzer and hydrogen tank.

### Sensitivity analysis

The LCOM of the green methanol plant can be affected by various parameters, mainly including the methanol plant scale, variable feedstock or byproduct price, the capital cost of major equipment, and external policy factors such as carbon taxing. Therefore, in this section, we studied the effect of these key parameters the LCOM of green methanol plant with oxygen selling.

#### Effect of methanol plant scale on the LCOM

Fig. 8 presents the LCOM of green methanol plants for 10 kt/yr and 100 kt/yr, respectively. It can be found that the effect of plant scale on the LCOM is not significant. As the scale of the green methanol plant increases from 10 kt/yr to 100 kt/yr, the LCOM decreases by just 5–7% for these systems. The decline in the LCOM is mainly caused by the effect of economics of scale of the downstream CO<sub>2</sub> hydrogenation unit. The LCOM from the downstream CO<sub>2</sub> hydrogenation unit declines by 40% as the plant size increases tenfold. However, the cost from CO<sub>2</sub>

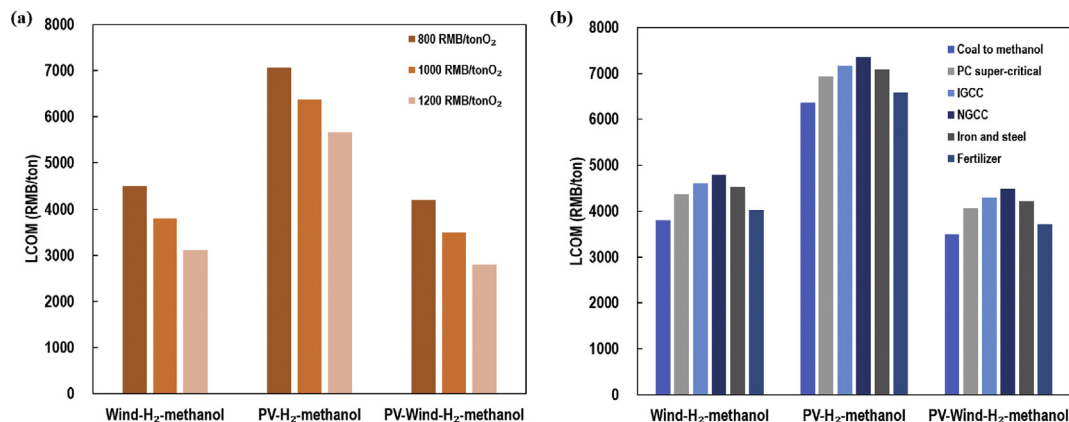


Fig. 9 – Effect of (a) oxygen selling cost and (b) CO<sub>2</sub> capture source on the LCOM of green methanol plant.

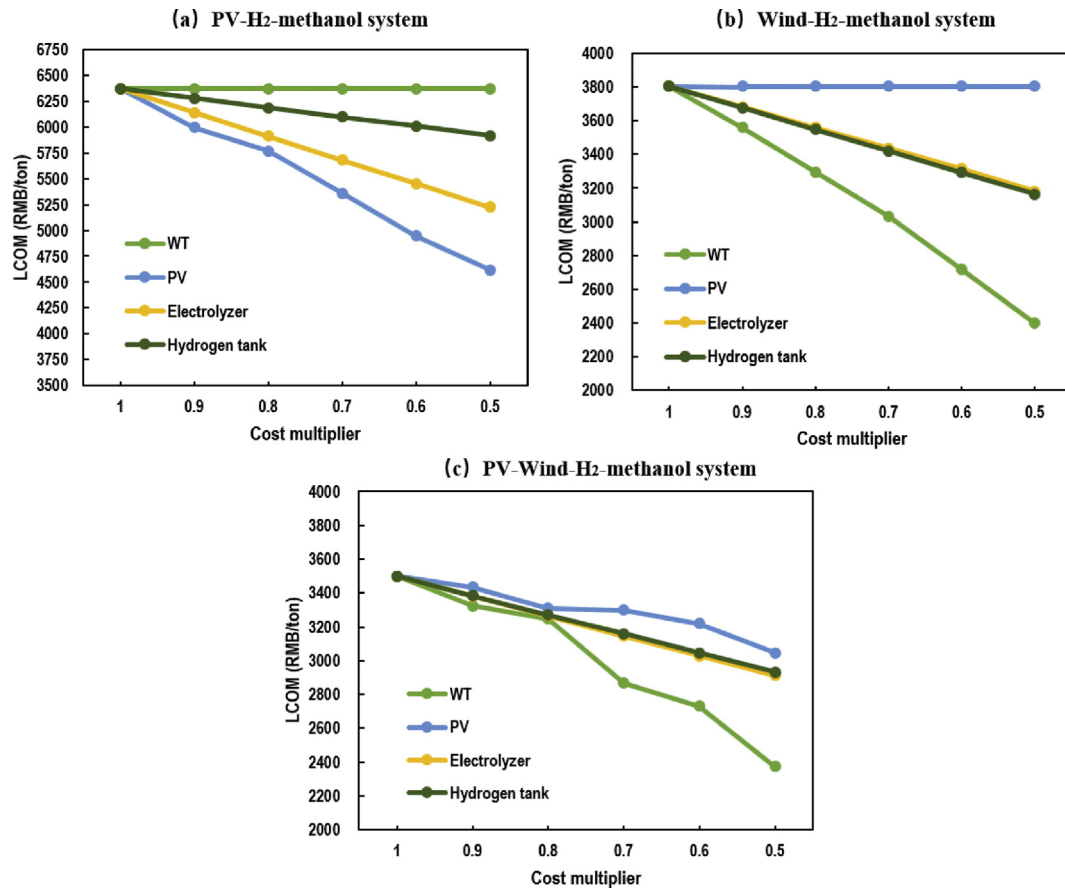


Fig. 10 – Single-factor effect of variable parameters on the LCOM.

hydrogenation unit just occupies less than 25% of the total cost in all cases. In contrast, the effect of increased plant scale on the LCOM from the upstream green hydrogen-electricity production system is not obvious partly because when we optimize the upstream system, the unit cost of each component was set fixed based on the average cost of wind turbines and PV modules of commercial wind farms or PV plants.

#### Effect of CO<sub>2</sub> source and oxygen selling price on the LCOM

The CO<sub>2</sub> capture cost and oxygen selling price were investigated to see how the O&M cost influences the LCOM based on the 10 kt/yr green methanol plant. The cost of oxygen selling price was set as 800, 1000 and 1200 RMB/ton. Note that when the sensitivity of O<sub>2</sub> price is studied, the CO<sub>2</sub> cost is 50 RMB/ton. As shown in Fig. 9(a), selling by-product oxygen has a great impact on the methanol production cost. An increase of 200 RMB/ton in the selling price of oxygen can reduce the methanol cost by 697 RMB/ton.

The CO<sub>2</sub> cost is highly dependent on its capture source. In the baseline case, we assumed the CO<sub>2</sub> captured from a conventional coal-based methanol plant. In this section, CO<sub>2</sub> captured from PC super-critical, IGCC, NGCC power plant, fertilizer and iron-making industry was also considered. The CO<sub>2</sub> cost of each scenario used in this study is listed in Table 2 with reference to literature [55]. When the sensitivity of CO<sub>2</sub> cost is studied, the selling price of oxygen is 1000 RMB/ton. As

shown in Fig. 9(b), the CO<sub>2</sub> source has an apparent effect on the LCOM of green methanol plant. For PV-Wind-H<sub>2</sub>-methanol system, the LCOM ranges between 3498 and 4486 RMB/ton according to various capture sources. Therefore, choosing a more economic CO<sub>2</sub> sources is a feasible way to get better economy of the green methanol plant.

#### Sensitivity analysis of the cost of system components

The cost of the green methanol plant is most affected by the capital investment of the upstream hybrid energy systems which is mainly sensitive to the cost of wind turbines, PV modules, PEM electrolyzer and hydrogen storage. Fig. 10(a–c) presents the single-factor effect of wind turbines, PV modules, PEM electrolyzer and hydrogen storage cost on the LCOM of the 10 kt/yr green methanol plant with different upstream hybrid energy systems. A cost multiplier set between 0.5 and 1 is introduced to determine the cost range of these components, which denotes the capital cost adopted dividing its baseline cost.

The one-way sensitivity analysis results in Fig. 10 was obtained by varying one factor with other factors fixed. In general, the LCOM for green methanol plant decreases when the cost of one component goes down. For PV-H<sub>2</sub>-methanol system, its LCOM is most sensitive to the PV cost, followed by the cost of electrolyzer. When the PV cost decreases by 50%, the LCOM of the PV-H<sub>2</sub>-methanol system is 4618 RMB/ton,



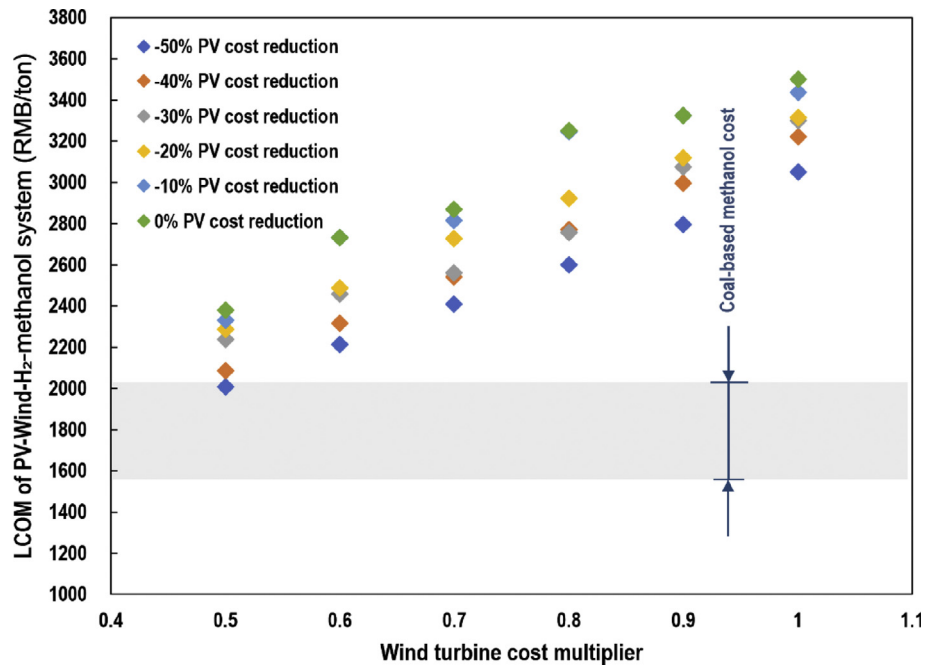


Fig. 11 – Effect of wind turbine and PV cost on the LCOM of PV-Wind-H<sub>2</sub>-methanol system.

decreased by 28%. For PV-Wind-H<sub>2</sub>-methanol and Wind-H<sub>2</sub>-methanol system, the most influential economic factor is the wind turbine cost. As shown in Fig. 10(b)–(c), when the cost of wind turbines drops by 50% from current price, the LCOM of the PV-Wind-H<sub>2</sub>-methanol and Wind-H<sub>2</sub>-methanol system decreases by 32% and 37%, respectively. Under above scenario, the LCOM of the PV-Wind-H<sub>2</sub>-methanol and Wind-H<sub>2</sub>-methanol system can reach 2375 and 2400 RMB/ton,

respectively, which is still higher than the conventional coal-based methanol production cost [1,44].

In fact, the cost of all these components will go down in the future, so the double-factor effect of wind turbine and PV module cost on the LCOM of the PV-Wind-H<sub>2</sub>-methanol system was also analyzed because the two components are predicted to have great cost reduction potential [65–67]. As shown in Fig. 11, to make the LCOM of PV-Wind-H<sub>2</sub>-methanol

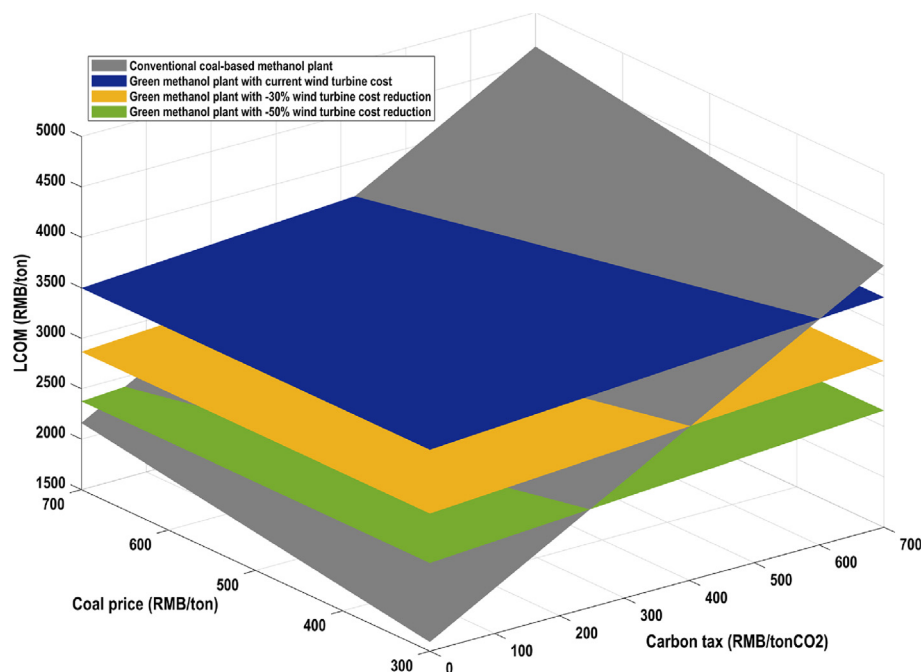


Fig. 12 – Effect of carbon tax on LCOM of the PV-Wind-H<sub>2</sub>-methanol system.

system to lower than that of a typical coal-based methanol plant estimated based on coal price (300–600 RMB/ton) in Ordos [68] (Economic summary of the coal-based methanol plant is available in the supplement material), the cost of wind turbines and PV modules need to be reduced by nearly 50% at the same time, which is not easy to achieve in the near future. Therefore, to make the green MeOH plant economically competitive as soon as possible, not only the cost of components needs to be reduced, but also other factors such as external policies need to be resorted to.

#### Sensitivity analysis of carbon tax

The unit carbon emission of the green methanol plant is 0.4 tonCO<sub>2</sub>/ton-MeOH, 80% of which is indirect carbon emission and the other 20% is direct emission. Direct emission mainly comes from the combustion of purge gas while indirect emission from consumption of utilities such as steam and cooling water. The unit carbon emission of the green methanol plant is approximately 90% less than that of the conventional coal-based methanol plant (3.57 tonCO<sub>2</sub>/ton-MeOH) referenced from our previous studies [1]. To explore the possibility of the green methanol plant being economically competitive with conventional methanol plant, carbon taxing is considered in this section. The impact of carbon tax on the LCOM was analyzed considering the downward trend of wind turbine cost since it is the most influential economic factor among the system components. The carbon tax range was set according to the level suggested by World Bank which reported a carbon tax that complies with the Paris agreement's temperature targets should reach a level of 40–80\$/ton CO<sub>2</sub> in 2020 and 50–100\$/ton CO<sub>2</sub> in 2030 [69]. Two additional scenarios where the cost of wind turbines would fall by 30% and 50% were analyzed, which were set according to previous projections on wind power cost reduction. A 2016 study funded by the DOE anticipated cost reductions ranged from 24% to 30% by 2030 and can reach 40% as research and development programs and technology learning lead to additional efficiencies; National Renewable Energy Laboratory (NREL) predicts a “SMART” wind plant of the future will be able to achieve even further cost reductions - up to 50% by 2030 [67].

As shown in Fig. 12 under the current cost of wind turbines, it's possible to make the green methanol plant to be economically competitive with conventional methanol plant by applying carbon tax. Under the current wind turbine cost and 500 RMB/ton coal price, when the carbon tax is higher than 512 RMB/tonCO<sub>2</sub>, the PV-Wind-H<sub>2</sub>-methanol system will be able to compete with the conventional methanol plant. If the cost of wind turbines drops by 30% as predicted, to make the LCOM of the PV-Wind-H<sub>2</sub>-methanol system equal to that of conventional methanol plant, the carbon tax only needs to be higher than 312 RMB/tonCO<sub>2</sub>. Under more aggressive circumstances in which the cost of wind turbines falls by 50%, the minimum carbon tax can be reduced to 157 RMB/tonCO<sub>2</sub>. In general, as the cost of wind turbines decreases more and coal prices increase, the minimum value of carbon tax to make the LCOM of green methanol plant equal to that of the conventional methanol plant will be smaller.

## Conclusion

In this paper, we conducted an integrative techno-economic analysis on the green methanol plant, considering optimal design of upstream renewable hydrogen-electricity production. The economic results of the green methanol plant with different upstream hybrid energy configurations show that for the analyzed site (Ordos, China), the green methanol plant adopting PV-Wind-H<sub>2</sub> system with oxygen selling has the lowest LCOM (3498 RMB/ton-MeOH), which is, however, about 86% higher than conventional coal-based methanol cost. In terms of the cost structure, the capital cost is the major economic contributor for all three cases, more than 90% of which comes from the upstream hybrid energy system.

The sensitivity analysis results indicate that 1) the effect of economics of scale of the analyzed system is not obvious in this study; 2) oxygen selling option and the CO<sub>2</sub> source has a significant influence on the O&M cost; 3) for PV-Wind-H<sub>2</sub>-methanol and Wind-H<sub>2</sub>-methanol system, the most influential economic factor is the wind turbine cost. When the cost of wind turbines drops by 50% from current price, the LCOM of the PV-Wind-H<sub>2</sub>-methanol and Wind-H<sub>2</sub>-methanol system decreases by 32% and 37%, respectively; 4) if considering carbon tax, it's possible to make the green methanol plant to be economically competitive with conventional methanol plant. Under the current wind turbine cost and 500 RMB/ton coal price, when the carbon tax is higher than 512 RMB/tonCO<sub>2</sub>, the PV-Wind-H<sub>2</sub>-methanol system will be able to compete with the conventional methanol plant; If the cost of the wind turbine is further reduced, the carbon tax that makes the green methanol plant economical will be much lower.

## Declaration of competing interest

The authors declare no competing interests.

## Acknowledgments

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## Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.ijhydene.2021.11.148>.

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